

Volatility-Gated Motion on a SPY/TLT Efficient Frontier

Harness Research

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Abstract

This research note tests a causal allocation rule that begins at a rolling mean-variance growth optimum and moves along the estimated SPY/TLT efficient frontier. A five-session volatility surprise determines direction, while the absolute growth gap versus a static 60/40 SPY/TLT reference determines amplitude. Sinusoidal, linear, and quadratic causal interpolators are compared from January 2023 onward after proportional trading costs.

Disclaimer. This document is research software output, not investment advice, and not evidence of future returns.

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1 Thesis and research design

This note formalises an investment thesis as a testable mathematical object and evaluates it on historical data. The pipeline is deliberately mechanical: a strategy maps the price history available at time t to target weights (Section 2); a backtest converts those weights into a net-of-cost return series; and every number quoted in the text is generated from that series, not typed by hand (Sections 3–5 define the estimators; Section 10 reports them).

The research sample contains 930 sessions from 2023-01-03 to 2026-09-18. Trading costs are charged at 5 basis points per unit of turnover. These assumptions are fixed before the results are read, and they determine what the figures can and cannot show.

The key design rule is *no look-ahead*: the weight vector applied to the return of session t may depend only on information dated strictly before that return is realised. Formally, if $\mathcal{F}_{t-1}$ is the information set at the close of session $t - 1$, then

$$w_t = g(\mathcal{F}_{t-1}), \quad r_t^{\text{strat}} = \sum_i w_{i,t} r_{i,t} - c \sum_i |w_{i,t} - w_{i,t}^{\text{drift}}|, \quad (1)$$

where $w_{i,t}^{\text{drift}}$ is the previous target weight after drifting with the market and $c = \text{CostBps}/10^4$ is the proportional cost rate.

1.1 Hypotheses

- H1 The sinusoidally interpolated rule has a higher net Sharpe ratio than static 60/40 SPY/TLT.
- H2 A positive short-horizon volatility surprise reduces the strategy's frontier risk coordinate.
- H3 The ranking of sine, linear, and quadratic interpolation is not invariant to transaction costs.

2 Portfolio accounting

2.1 Currency conversion

Let asset i be quoted in currency Q_i and let the account base currency be B . If X_t^{B,Q_i} denotes units of Q_i per one unit of B , a native price $P_{i,t}^{Q_i}$ has base-currency value

$$P_{i,t}^B = \frac{P_{i,t}^{Q_i}}{X_t^{B,Q_i}}. \quad (2)$$

Equivalently, with a direct multiplier $Y_t^{Q_i,B} = \left(X_t^{B,Q_i}\right)^{-1}$ in units of B per unit of Q_i ,

$$P_{i,t}^B = P_{i,t}^{Q_i} Y_t^{Q_i,B}. \quad (3)$$

Forward-filling an FX series onto an asset's trading dates is an alignment convention, not an estimate of an unobserved intraday rate. When all assets share the base currency, both equations are the identity.

2.2 NAV and target weights

With cash C_t , position quantity $q_{i,t}$, and base-currency price $P_{i,t}^B$, net asset value (NAV) is

$$V_t = C_t + \sum_{i=1}^m q_{i,t} P_{i,t}^B. \quad (4)$$

For target weights $w_i \geq 0$ satisfying $\sum_i w_i \leq 1$, the ideal target quantity is

$$q_{i,t}^* = \frac{w_i V_t}{P_{i,t}^B}. \quad (5)$$

Equation (5) is the bridge between economic allocation and position size. Unallocated weight, $1 - \sum_i w_i$, remains in cash and earns zero. The backtest uses fractional quantities so that rounding does not obscure the mathematics.

3 Returns, wealth, and performance

For a consistently adjusted price series P_t , the one-period simple return is

$$r_t = \frac{P_t}{P_{t-1}} - 1, \quad (6)$$

and the log return is

$$\ell_t = \log\left(\frac{P_t}{P_{t-1}}\right) = \log(1 + r_t). \quad (7)$$

Simple returns aggregate across simultaneously held assets; log returns add through time.

Starting from wealth V_0 , compounded wealth after T returns is

$$V_T = V_0 \prod_{t=1}^T (1 + r_t). \quad (8)$$

Total return and compound annual growth rate over Y calendar years are

$$R_{\text{tot}} = \frac{V_T}{V_0} - 1, \quad \text{CAGR} = (1 + R_{\text{tot}})^{1/Y} - 1. \quad (9)$$

Using $A = 252$ sessions per year and a zero risk-free rate, annualised volatility and the Sharpe ratio are

$$\hat{\sigma}_{\text{ann}} = \hat{\sigma}_r \sqrt{A}, \quad \hat{S} = \frac{\bar{r}}{\hat{\sigma}_r} \sqrt{A}. \quad (10)$$

For the rolling graph, $\bar{r}$ and $\hat{\sigma}_r$ are recomputed over a trailing window using the sample standard deviation; the estimate is path-dependent and moves sharply when an extreme observation enters or leaves the window.

Let $M_t = \max_{0 \leq s \leq t} V_s$ be the running high-water mark. Drawdown and maximum drawdown are

$$D_t = \frac{V_t}{M_t} - 1, \quad D_{\text{max}} = \min_{0 \leq t \leq T} D_t. \quad (11)$$

These summaries describe one realised path. They do not quantify parameter uncertainty, multiple testing, tail severity beyond the observed sample, or the probability of future loss.

4 Estimation and uncertainty

4.1 Gaussian maximum likelihood

For returns $x_1, \dots, x_n$, assume as a working approximation that $x_j \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$. The log-likelihood, up to an additive constant, is

$$\mathcal{L}(\mu, \sigma^2) = -\frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} \sum_{j=1}^n (x_j - \mu)^2. \quad (12)$$

Maximisation yields

$$\hat{\mu} = \frac{1}{n} \sum_{j=1}^n x_j, \quad \hat{\sigma}^2 = \frac{1}{n} \sum_{j=1}^n (x_j - \hat{\mu})^2. \quad (13)$$

The divisor n is the maximum-likelihood convention, not the unbiased one. The estimated standard error of the mean is

$$\widehat{\text{se}}(\hat{\mu}) = \frac{\hat{\sigma}}{\sqrt{n}}. \quad (14)$$

For the daily strategy returns of this study, $\hat{\mu} = 4.030e-04$, $\hat{\sigma}^2 = 5.943e-05$ and $\widehat{\text{se}}(\hat{\mu}) = 2.528e-04$. An estimated edge is only interesting relative to this error: the statistic $\hat{\mu}/\widehat{\text{se}}(\hat{\mu})$ is a t -type ratio, not a significance test, because returns are neither established as Gaussian nor as independent.

4.2 Stationary bootstrap for the Sharpe ratio

Daily returns are serially dependent enough that the iid interval is unreliable. The stationary bootstrap of Politis and Romano resamples blocks with geometric random length. With mean block length $1/p$, define the index sequence recursively: I_1 is uniform on $\{1, \dots, n\}$, and

$$I_{k+1} = \begin{cases} (I_k \bmod n) + 1, & \text{with probability } 1 - p, \\ \text{uniform on } \{1, \dots, n\}, & \text{with probability } p. \end{cases} \quad (15)$$

For replication $b = 1, \dots, B$ (here $B = 1000$), form $x_k^{*b} = x_{I_k}$ for $k = 1, \dots, n$ and compute $\hat{S}^{*b}$ from Equation (10). The percentile interval for the Sharpe ratio is

$$\left[\hat{S}_{(\alpha/2)}^*, \hat{S}_{(1-\alpha/2)}^* \right], \quad (16)$$

the empirical $\alpha/2$ and $1-\alpha/2$ quantiles of the replications. For this study the interval is $[-0.16, 1.98]$ around the point estimate 0.83. The interval captures sampling noise under the resampling assumptions only; it does not correct for choosing the strategy after seeing data.

5 Position sizing

For a small risky return R and fraction f of wealth invested, a second-order expansion of expected log growth is

$$\mathbb{E}[\log(1 + fR)] \approx f \mathbb{E}[R] - \frac{f^2}{2} \mathbb{E}[R^2]. \quad (17)$$

When the mean is small relative to volatility, $\mathbb{E}[R^2] \approx \text{Var}(R)$, and maximising over f gives the familiar approximate optimiser

$$f^* \approx \frac{\mathbb{E}[R]}{\text{Var}(R)}. \quad (18)$$

Because $\mathbb{E}[R]$ and $\text{Var}(R)$ are estimated with error (Section 4), the full-Kelly fraction is systematically too aggressive. A fractional-Kelly rule scales it by $\lambda \in (0, 1]$ and imposes a cap $f_{\max}$, after subtracting the per-trade cost c :

$$f = \min\left(f_{\max}, \lambda \frac{\max(\hat{\mu} - c, 0)}{\hat{\sigma}^2}\right). \quad (19)$$

This is “Kelly-style”, not an exact Kelly solution: return distributions are not established, parameters are estimated, and the resulting weights are then constrained to $\sum_i w_i \leq 1$ as in Section 2. Growth is concave in f around f^* , so a fraction $\lambda < 1$ gives up a small amount of expected growth in exchange for a substantially smaller variance and drawdown.

For this study, plugging the estimates of Section 4 into Equation (18) gives an approximate full-Kelly fraction of 6.78; with $\lambda = 0.50$ and cap $f_{\max} = 1.00$, Equation (19) gives 1.00. A fraction above one would imply leverage; the strategy does not use these figures, which serve only to show how fragile the full-Kelly value is given the standard error above.

6 Thesis-specific model

The construction is a deliberately small two-asset implementation of the mean–variance idea in Markowitz (1952). Let $r_t = (r_{S,t}, r_{T,t})^\top$ denote the adjusted-close returns of SPY and TLT. At decision time t , only returns through t are used. With at most $L = 126$ observations, the rolling estimators are

$$\hat{\mu}_t = 252 \bar{r}_t, \quad \hat{\Sigma}_t = 252 \widehat{\text{Cov}}_t(r) + \epsilon I, \quad (20)$$

where the small ridge ϵI makes the numerical covariance positive definite without materially changing the frontier.

For $a \in [0.05, 0.95]$, define the fully invested portfolio $w(a) = (a, 1 - a)^\top$. The long-only global minimum-variance point and the mean–variance growth anchor are

$$a_t^{\text{gmV}} = \arg \min_a w(a)^\top \hat{\Sigma}_t w(a), \quad a_t^* = \arg \max_{a \in \mathcal{F}_t} \left\{ w(a)^\top \hat{\mu}_t - \frac{\gamma}{2} w(a)^\top \hat{\Sigma}_t w(a) \right\}, \quad (21)$$

with $\gamma = 3$. The set $\mathcal{F}_t$ is the estimated efficient segment from the GMV portfolio to the admissible endpoint having the higher estimated mean. Because there are two assets, moving along this segment is equivalent to changing the SPY coordinate a .

The user’s proposed “risk-free” 60/40 split is instead denoted $b = (0.60, 0.40)^\top$ and treated as a *risky reference portfolio*. Calling it risk-free would be mathematically incorrect: its return is stochastic and can be negative. Over the latest five completed sessions, define its compound-return gap from the anchor as

$$d_t = \prod_{j=0}^4 \left(1 + w(a_t^*)^\top r_{t-j} \right) - \prod_{j=0}^4 \left(1 + b^\top r_{t-j} \right). \quad (22)$$

Thus the rule has exactly zero displacement when the two five-session growth rates agree.

Let $\hat{\sigma}_{5,t}$ and $\hat{\sigma}_{21,t}$ be annualised sample volatilities of $b^\top r$ over five and 21 sessions. The signed raw frontier coordinate is

$$q_t = \text{clip}\left(\frac{\hat{\sigma}_{5,t}/\hat{\sigma}_{21,t} - 1}{0.50}, -1, 1\right) \min\left\{\frac{|d_t|}{\hat{\sigma}_{21,t}\sqrt{5/252}}, 1\right\}. \quad (23)$$

Positive q_t means the recent volatility estimate has risen and therefore calls for less risk; negative q_t calls for more risk. The absolute growth gap is a throttle, not a directional return forecast.

For $n = 5$, the causal interpolated coordinate is

$$\tilde{q}_t^{(k)} = \sum_{j=1}^n \kappa_j^{(k)} q_{t-n+j}, \quad \kappa_j^{(k)} = \frac{h_k(j)}{\sum_{\ell=1}^n h_k(\ell)}, \quad (24)$$

where $h_{\text{sine}}(j) = \sin(\pi j/(2n))$, $h_{\text{linear}}(j) = j$, and $h_{\text{poly}}(j) = j^2$. All three filters are one-sided; none uses a centred window or a future observation. Finally, with a_t^{hi} denoting the higher-risk endpoint of $\mathcal{F}_t$,

$$a_{t+1} = \begin{cases} a_t^* + \tilde{q}_t(a_t^{\text{gm}} - a_t^*), & \tilde{q}_t \geq 0, \\ a_t^* + (-\tilde{q}_t)(a_t^{\text{hi}} - a_t^*), & \tilde{q}_t < 0. \end{cases} \quad (25)$$

The target chosen at the close of t earns the return from t to $t + 1$, which is the no-lookahead convention enforced by the harness.

7 Interpolation comparison

Table 1 reports all three filters on the same observations, parameters, and five-basis-point turnover charge. Static 60/40 is included because it is the economic reference; SPY remains the generic harness benchmark in Table 2.

Table 1: Net performance by interpolation rule.

Rule	CAGR	Volatility	Sharpe	Max drawdown	Avg. turnover
Sine	9.86%	12.24%	0.83	-16.83%	7.56%
Linear	9.77%	12.24%	0.82	-16.87%	7.68%
Quadratic	9.57%	12.22%	0.81	-16.87%	8.08%
Static 60/40	12.32%	11.19%	1.09	-12.72%	—

The pre-registered outcomes are: H1 **Not supported**, H2 **Supported by construction**, and H3 **Not supported**. “Supported” describes only this historical path and is not a claim of population-level superiority.

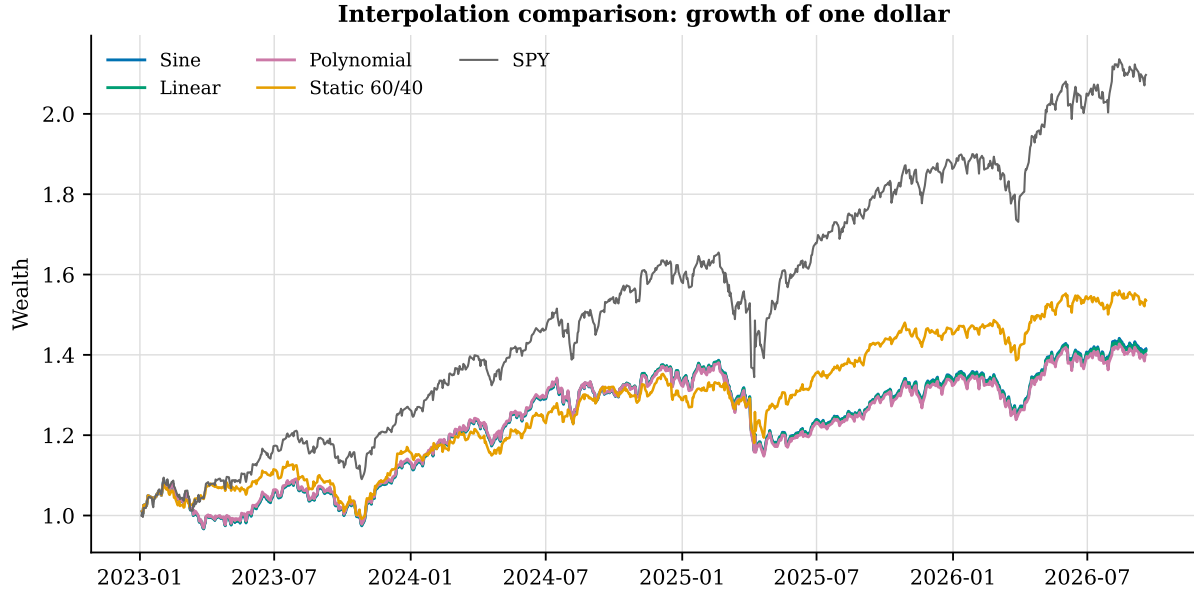


Figure 1: Cumulative wealth for the three interpolators, static 60/40, and SPY.

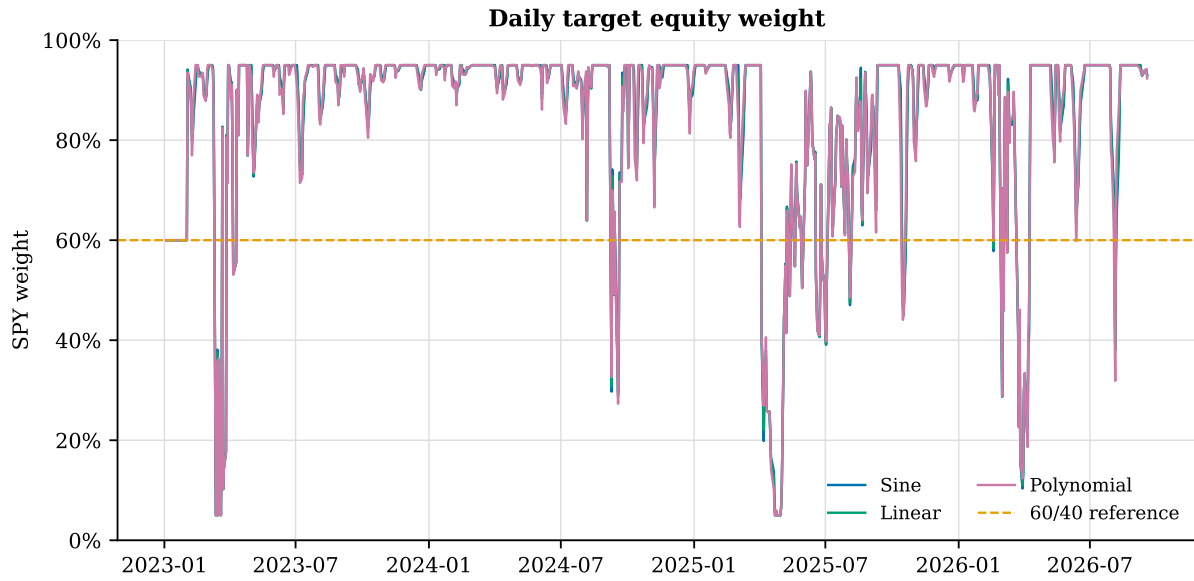


Figure 2: SPY target weights. Differences between filters are isolated to the five-session kernel.

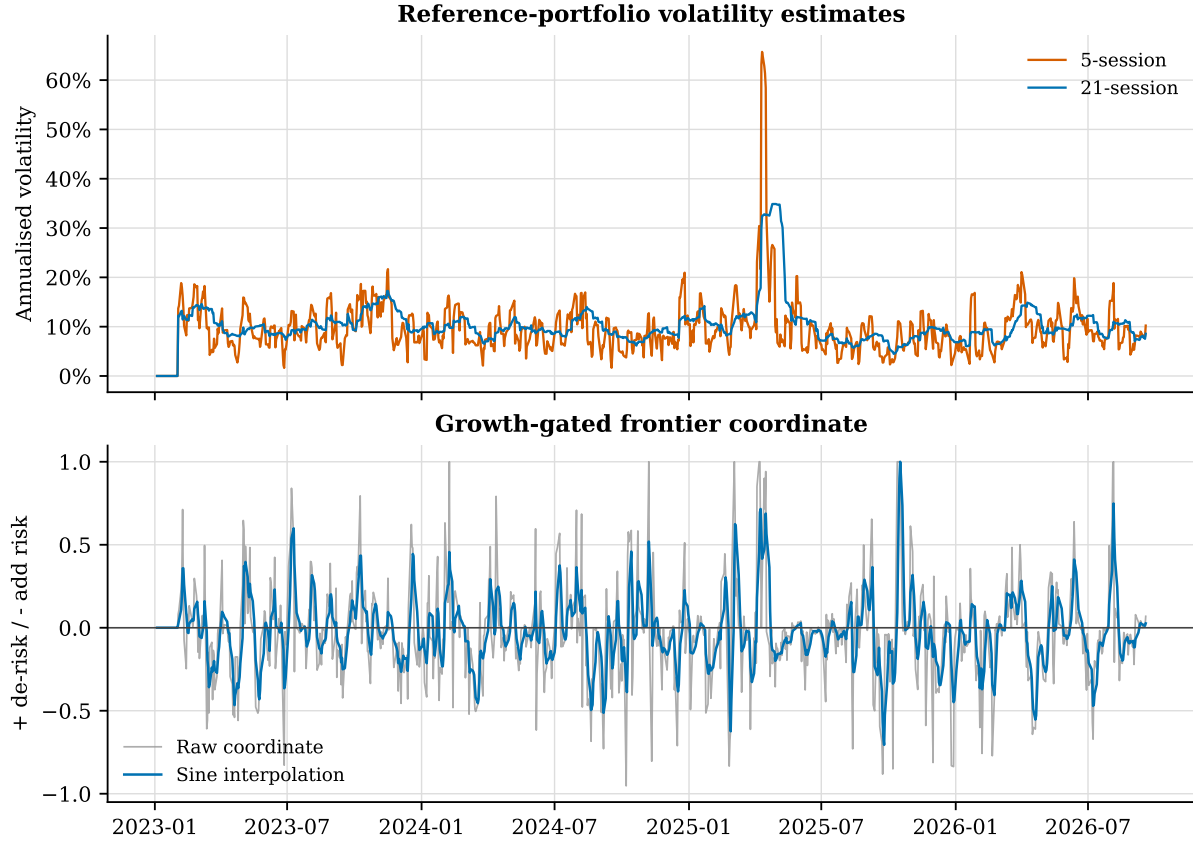


Figure 3: Short- and long-window reference volatility (top) and raw versus sine-smoothed frontier coordinate (bottom). Positive coordinates de-risk.

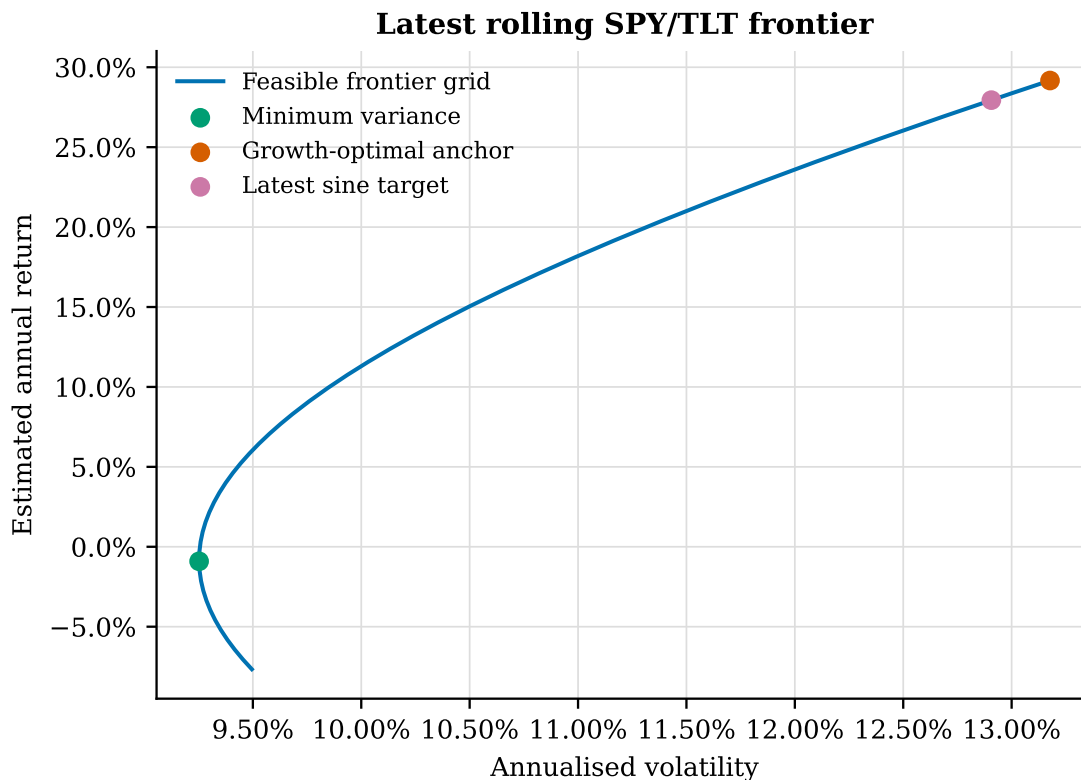


Figure 4: Latest estimated frontier, anchor, minimum-variance point, and sine target. This is a rolling estimate, not a stable structural frontier.

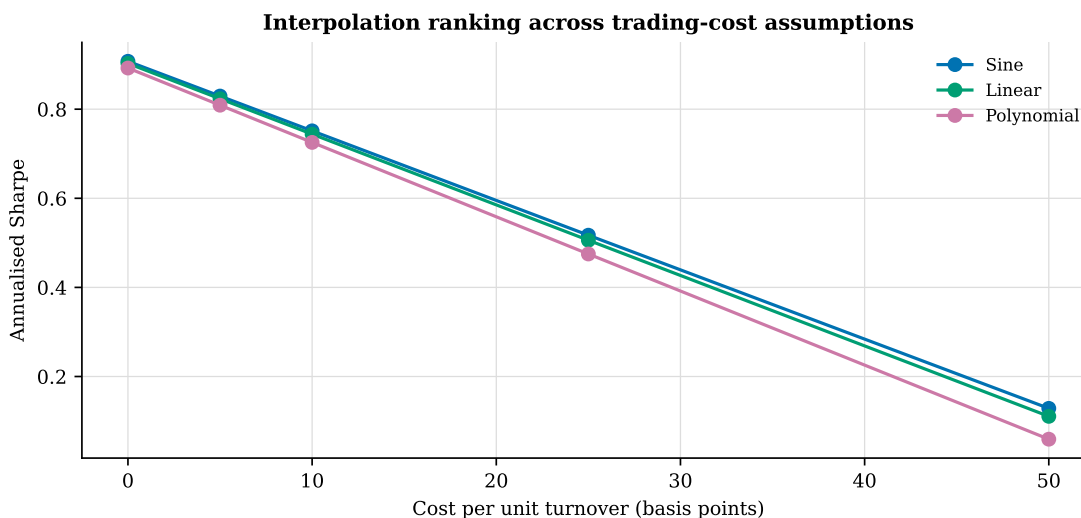


Figure 5: Net Sharpe ratios under alternative proportional cost assumptions.

8 Trade and broker interpretation

The latest completed-session sine target, dated 2026-09-18, is 91.77% SPY and 8.23% TLT; its smoothed coordinate is 0.04. These are model weights, not a recommendation or an executable

order. Exact share quantities depend on current IBKR paper-account NAV, positions, FX rates, and prices. The supplied planner reads those values, sizes whole-share differences, marks every order `transmit=False`, and never calls the broker’s placement method. The full historical target changes are in `model_trade_ledger.csv`.

Adjusted Yahoo closes are used for the performance study because they represent distributions and splits. IBKR TRADES bars are execution prices rather than a total-return series; swapping them into this study would silently change the return definition. IBKR is therefore used only for paper-account contract validation and order planning.

9 References

- H. Markowitz (1952), “Portfolio Selection,” *Journal of Finance* 7(1), 77–91, doi:10.1111/j.1540-6261.1952.tb01525.x.
- W. F. Sharpe (1994), “The Sharpe Ratio,” *Journal of Portfolio Management* 21(1), 49–58, author-hosted version.
- D. N. Politis and J. P. Romano (1994), “The Stationary Bootstrap,” *JASA* 89(428), 1303–1313, doi:10.1080/01621459.1994.10476870.

10 Results

All values below are generated from the backtest output; none is typed by hand. The sample spans 930 sessions from 2023-01-03 to 2026-09-18 with costs of 5 basis points.

Table 2: Performance summary (net of costs). The interval for the strategy Sharpe ratio is the stationary-bootstrap percentile interval of Equation (16).

	Strategy	Benchmark
CAGR	9.86%	22.22%
Sharpe ratio	0.83	1.42
Maximum drawdown	-16.83%	-18.76%
Total return	41.49%	
Annualised volatility	12.24%	
Sharpe 95% interval	[−0.16, 1.98]	

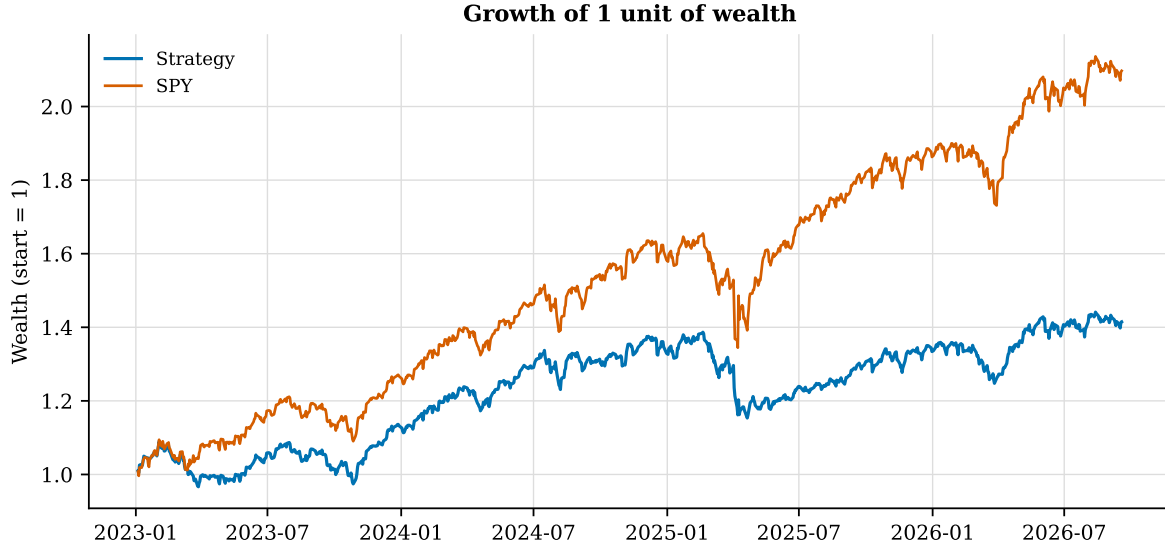


Figure 6: Cumulative wealth of the strategy and its benchmark (Equation (8)).

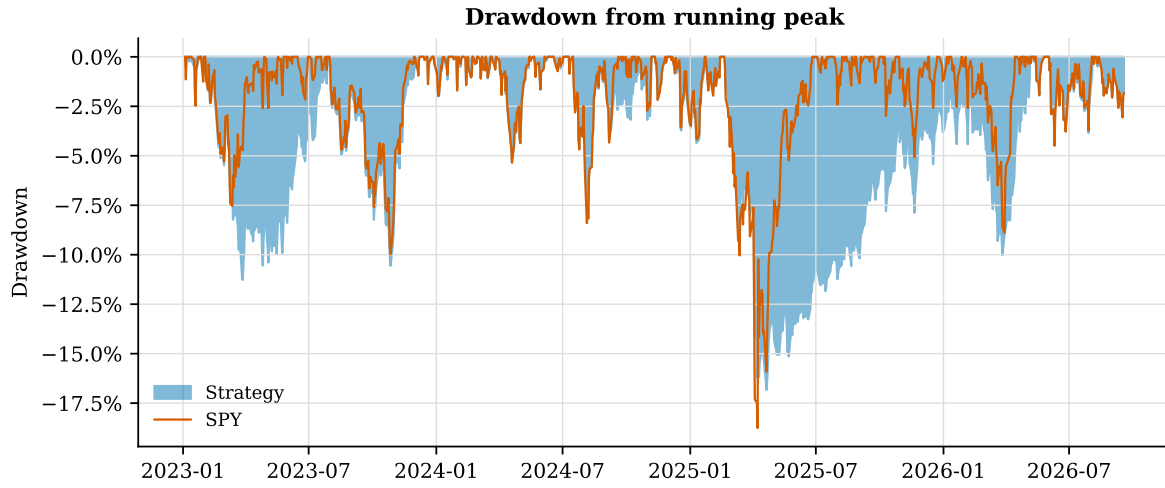


Figure 7: Drawdown from the running peak (Equation (11)).

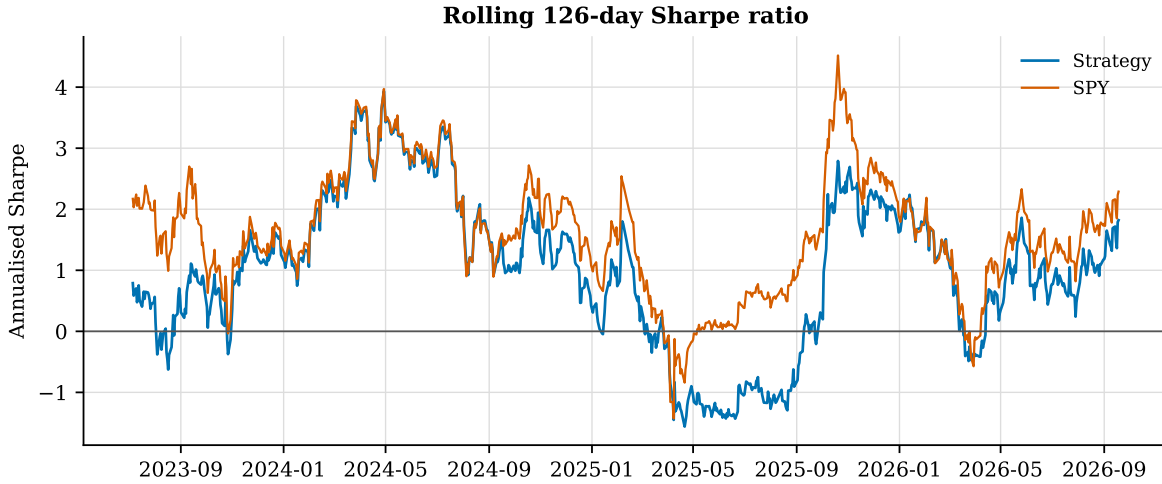


Figure 8: Rolling annualised Sharpe ratio (Equation (10)). Adjacent windows overlap almost completely, so the path is descriptive rather than a sample of independent estimates.

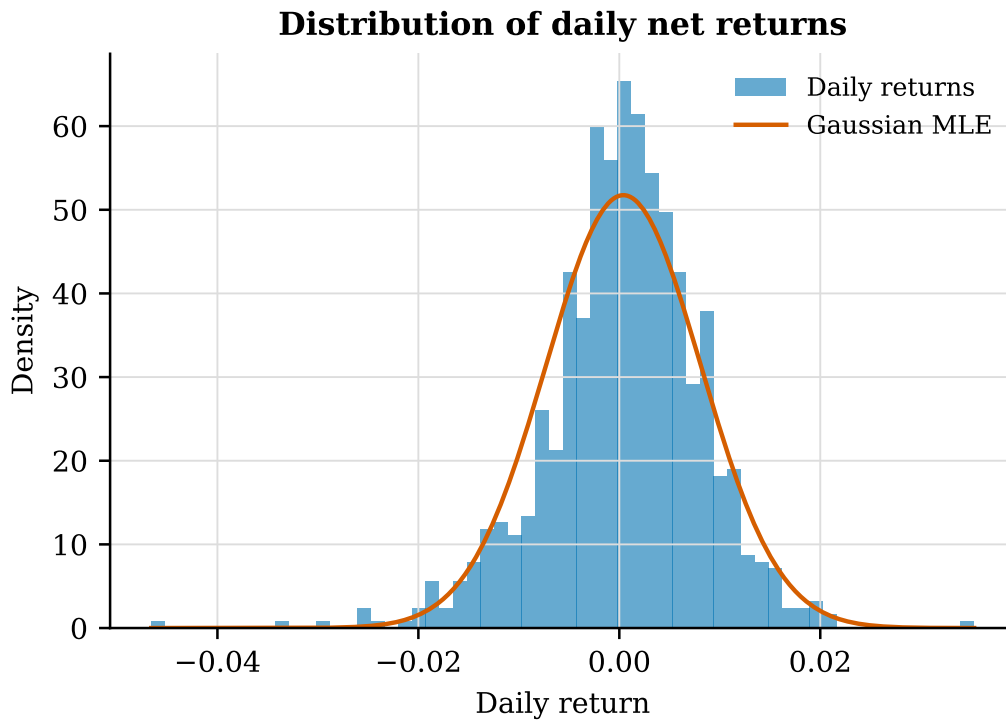


Figure 9: Distribution of daily strategy returns. Fat tails and skew, if visible, contradict the Gaussian working assumption of Equation (12).

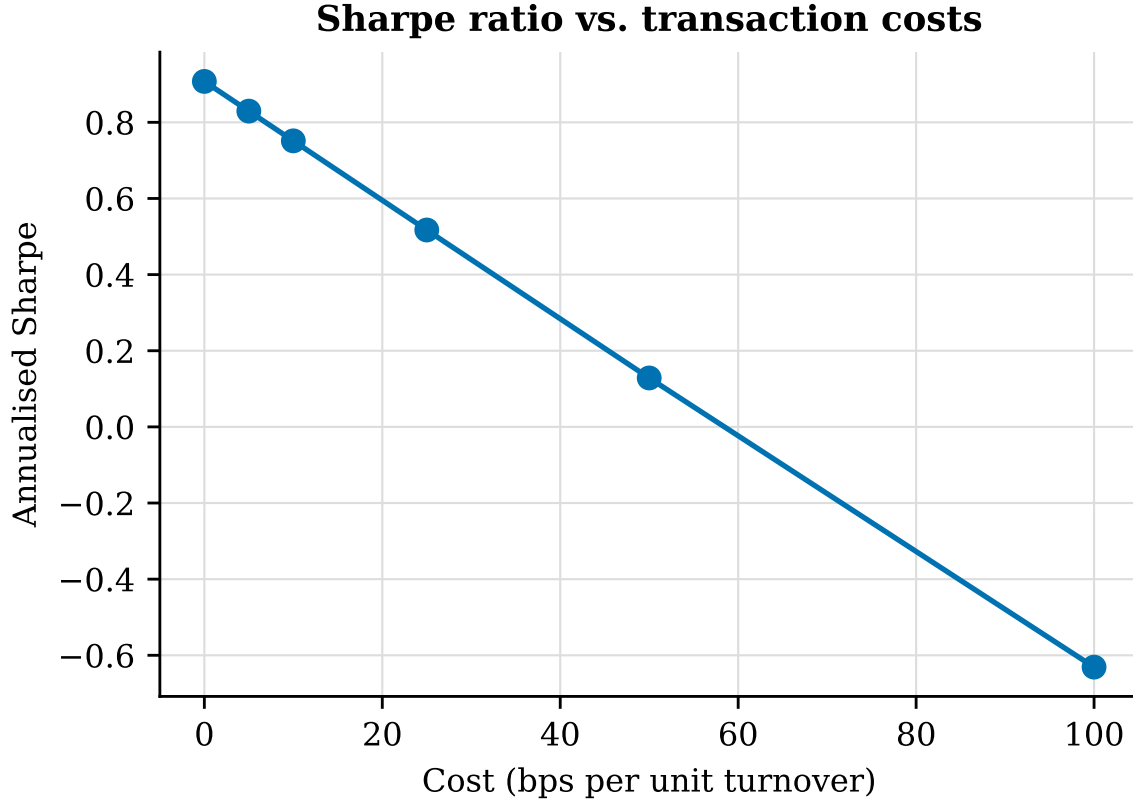


Figure 10: Sensitivity of performance to the transaction-cost assumption. A result that only survives at low cost is fragile.

11 Evidence and limitations

The figures show one realised path and are not evidence of dominance. The following caveats apply to every paper produced by this pipeline, whatever the numbers say.

- **Sample length.** The sample has 930 sessions (2023-01-03 to 2026-09-18). The standard error of a Sharpe ratio estimated from T years of data is roughly $\sqrt{(1 + S^2/2)/T}$, so short samples cannot distinguish skill from luck; see the interval $[-0.16, 1.98]$.
- **Costs.** Costs are a proportional 5 basis points per unit of turnover. Real costs include spreads that widen in stress, market impact, commissions, financing, and taxes, none of which are modelled.
- **Overfitting and multiple testing.** The thesis was formulated with knowledge of history. Every parameter choice, ticker choice, and discarded variant increases the chance that the best-looking result is noise. The bootstrap interval does not account for this selection.
- **Model assumptions.** Gaussian and iid assumptions are working approximations. Financial returns are heteroskedastic, skewed, and heavy-tailed.
- **Data.** Adjusted closes are assumed correct and free of survivorship bias; the risk-free rate is taken as zero, and rankings can change with other choices.

- **No out-of-sample test.** Unless stated otherwise, no untouched holdout period was reserved.

This document is not investment advice. It is an automatically generated research note; nothing in it has been validated for live capital, and past performance says little about future returns.

Conclusion

The chain of argument is mechanical: consistent prices give returns (Equation (6)); weights that use only past information give a net-of-cost strategy return (Equation (1)); compounding gives wealth, growth, and drawdown (Equations (8)–(11)); and estimation theory gives an honest statement of uncertainty (Equation (16)). Over 2023-01-03 to 2026-09-18 the strategy delivered a CAGR of 9.86% and a Sharpe ratio of 0.83 against 22.22% and 1.42 for the benchmark, with a maximum drawdown of -16.83%. Given the limitations of Section 11, these are hypotheses for further out-of-sample study, not conclusions about the future.